

How Capable Is Qwen3.5 at OCR?

A Look at English and Arabic Book Scans

1 Introduction

I wanted to find out how far I could push Qwen3.5 as an OCR reader — not because it’s built for that job, but because it’s a vision-language model I could run entirely on my own machine, and I was curious whether that would be good enough. My plan was simple: feed it scanned pages from a few different books, in two different scripts, and check its transcription word-for-word against a text I’d verified by hand.

Getting even this far took some working around. I picked Qwen3.5 over Qwen2 because it’s simply the stronger model — a newer generation with better reasoning and multimodal ability — and it was still something my machine could run locally at a moderate pace. That didn’t make it fast: even with a capable-enough setup, individual pages still took several minutes to process. It was a balance between running the strongest model I reasonably could and still getting results back in a workable amount of time.

The test set: two different scans/editions of Aristotle’s *Nicomachean Ethics* in English, and one Arabic-language biographical work, *فريدة الدهر في طبقات قراء مصر* (Volume 1). Every OCR result below is checked against a hand-verified transcription of the same page.

2 How Accuracy Is Measured

Each result is scored three ways:

- **Word Error Rate (WER)** — out of all the words in the correct text, what percentage don’t line up with what the model produced? This is the most intuitive number: it’s roughly “how many words would I have to fix by hand.” Lower is better.
- **Character Error Rate (CER)** — the same idea, but measured letter by letter instead of word by word. In principle it’s a finer-grained version of WER.
- **Overall Similarity** — a general string-similarity ratio between the OCR output and the correct text.

One warning about CER and Similarity, which shows up repeatedly below: both are computed by lining up the two texts character-by-character, and that alignment can fall apart badly when just a few words get reordered or reworded — even if the surrounding text is almost perfect. So a page can score a rough CER while still being highly readable and mostly correct. WER is the number I trust most in this report; CER and Similarity are included for completeness, but should be read with that caveat in mind.

BOOK I.

EVERY art and every scientific inquiry, and similarly every action and purpose, may be said to aim at some good. Hence the good has been well defined as that at which all things aim. But it is clear that there is a difference in the ends; for the ends are sometimes activities, and sometimes results beyond the mere activities. Also, where there are certain ends beyond the actions, the results are naturally superior to the activities.

As there are various actions, arts, and sciences, it follows that the ends are also various. Thus health is the end of medicine, a vessel of shipbuilding, victory of strategy, and wealth of domestic economy. It often happens that there are a number of such arts or sciences which fall under a single faculty, as the art of making bridles, and all such other arts as make

the instruments of horsemanship, under horsemanship, and this again as well as every military action under strategy, and in the same way other arts or sciences under other faculties. But in all these cases the ends of the architectonic arts or sciences, whatever they may be, are more desirable than those of the subordinate arts or sciences, as it is for the sake of the former that the latter are themselves sought after. It makes no difference to the argument whether the activities themselves are the ends of the actions, or something else beyond the activities as in the above mentioned sciences.

If it is true that in the sphere of action there is an end which we wish for its own sake, and for the sake of which we wish everything else, and that we do not desire all things for the sake of something else (for, if that is so, the process will go on *ad infinitum*, and our desire will be idle and futile) it is clear that this will be the good or the supreme good. Does it not follow then that the knowledge of this supreme good is of great importance for the conduct of life, and that, *if we know it*, we shall be like archers who have a mark at which to aim, we shall have a better chance of attaining what we want? But, if this is the case, we must endeavour to comprehend, at least in outline, its nature, and the science or faculty to which it belongs.

It would seem that this is the most authoritative or architectonic science or faculty, and such is evidently the political; for it is the political science or faculty which determines what sciences are necessary in states, and what kind of sciences should be learnt,

and how far they should be learnt by particular people. We perceive too that the faculties which are held in the highest esteem, e.g. strategy, domestic economy, and rhetoric, are subordinate to it. But as it makes use of the other practical sciences, and also legislates upon the things to be done and the things to be left undone, it follows that its end will comprehend the ends of all the other sciences, and will therefore be the true good of mankind. For although the good of an individual is identical with the good of a state, yet the good of the state, whether in attainment or in preservation, is evidently greater and more perfect. For while in an individual by himself it is something to be thankful for, it is nobler and more divine in a nation or state.

Figure 1: Aristotle, Edition 1 — an older, archaic-style translation.

Metric	Value
Word Error Rate	11.99%
Character Error Rate	65.81%
Overall Similarity	33.42%

About 88% of words came back correct. Most of what’s wrong is trivial:

- **Small word swaps:** “at” read as “with”; “Is” read as “Does.” Wrong, but harmless.
- **Line-break hyphenation:** words split across a line break came through with a stray space, e.g. “subordinate” as “subordi- nate,” “comprehend,” as “com- prehend,”.

But a few errors go further than simple misreading — the model rewrites entire clauses into more modern, fluent phrasing instead of transcribing what’s actually printed:

- The original reads “...it must of the ends of all the other sciences...” — an awkward, archaic construction. Qwen’s output instead reads “...it follows that its end will comprehend the ends of all the other sciences...” — smoother, but not what’s on the page.
- The original reads “...it is noble to achieve it for a nation or state.” Qwen’s output reads “...it is nobler and more divine in a nation or state” — which happens to be near-identical to how the *modern* translation (Edition 2, below) phrases the same idea. The model appears to be smoothing archaic phrasing toward more familiar modern English rather than reading literally.

This is a useful early signal: even where Qwen's transcription looks fluent and readable, it isn't always faithful to the source.

4 Aristotle — Edition 2 (Modern Translation)

CHAPTER ONE

Every art and every inquiry, and similarly every action as well as choice, is held to aim at some good.¹ Hence people have nobly² declared that the good is that at which all things aim. But there appears to be a certain difference among the ends: some ends are activities, others are certain works apart from the activities themselves, and in those cases in which there are certain ends apart from the actions, the works are naturally better than the activities.³

Now, since there are many actions, arts, and sciences,⁴ the ends too are many: of medicine, the end is health; of shipbuilding, a ship; of generalship, victory; of household management, wealth. And in all things of this sort that fall under some one capacity⁵—for just as bridle making and such other arts as concern equestrian gear fall under horsemanship, while this art and every action related to warfare fall under generalship, so in the same manner, some arts fall under one capacity, others under another—in *all* of them, the ends of the architectonic ones are more choiceworthy than all those that fall under them, for these latter are pursued for the sake of the former. And it makes no difference at all whether the ends of the actions are the activities themselves or something else apart from these, as in the sciences mentioned.

CHAPTER TWO

If, therefore, there is some end of our actions that we wish for on account of itself, the rest being things we wish for on account of this end, and if we do not choose all things on account of something else—for in this way the process will go on infinitely such that the longing⁶ involved is empty and pointless—clearly this would be the good, that is, the best.⁷ And with a view to our life, then, is not the knowledge of this good of great weight, and would we not, like archers in possession of a target, better hit on what is needed? If this is so, then one must try to grasp, in outline at least, whatever it is and to which of the sciences or capacities it belongs.

But it might be held to belong to the most authoritative and most architectonic one,⁸ and such appears to be the political art.⁹ For it or-

dains what sciences there must be in cities and what kinds each person in turn must learn and up to what point. We also see that even the most honored capacities—for example, generalship, household management, rhetoric—fall under the political art. Because it makes use of the remaining¹⁰ sciences and, further, because it legislates what one ought to do and what to abstain from, its end would encompass those of the others, with the result that this would be the human good. For even if this is the same thing for an individual and a city, to secure and preserve the good of the city appears to be something greater and more complete: the good of the individual by himself is certainly desirable enough, but that of a nation and of cities is nobler and more divine.

The inquiry, then, aims at these things, since it is a sort of political inquiry.

Figure 2: Aristotle, Edition 2 — a modern translation, cleaner typesetting.

Metric	Value
Word Error Rate	3.17%
Character Error Rate	14.47%
Overall Similarity	84.79%

This is Qwen at its best, and the errors that remain are entirely cosmetic:

- **Footnote markers flattened:** “good.¹” came back as “good.1”; “longing⁶” came back as “longing⁶” — the model clearly recognized these as footnote markers, it just didn’t render them as true Unicode superscripts.
- **One real word swap:** “generalship,” read as “generals,” — a genuine misreading, but an isolated one.

No clauses were rewritten, no words were substituted with different meanings. The gap between this edition and Edition 1 — with the exact same model, the exact same task — is entirely attributable to scan/print quality. Clean, modern typesetting gets Qwen close to a perfect transcription; older or messier print does not.

5 Arabic — فريدة الدهر في طبقات قراء مصر, Volume 1

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

الحمد لله الذي هدانا إلى الإسلام، وفضلنا بالقرآن، وشرفنا بمحمد خير الأنام،
والصلاة والسلام على سيدنا محمد سيد البشر أجمعين، وعلى آله وأزواجه وأصحابه الطيبين
الطاهرين، وعلى التابعين ومن تبعهم بإحسان إلى يوم الدين، أما بعد:

فقد أدرك العقلاء والفضلاء أهمية علم التراجم وسير الأفراد، لهذا اعتنى به المتقدمون
والتأخرون، وذلك لأنه يتناول سير حياة الأعلام من الناس عبر العصور المختلفة، وسير
الأفراد من العلوم التي يحتاج إليها، إذ به يعرف الخلف أحوال السلف، وبه يُعرف الوفاء
ومحاسن الأخلاق، وسير الأولين جميلة، وأخبارهم جليلة، وأعمالهم عظيمة، وقد قصَّ الله
تعالى أخبار الأمم السالفة في أم الكتاب فقال تعالى: ﴿لَقَدْ كُنَّا فِي قَصَصِهِمْ عِبْرَةً لِأُولِي
الْأَلْبَابِ﴾ [يوسف: ١١١].

قال خاتمة المحققين محمد ابن الجزري في النشر:

«ومن نظر أسانيد كتب القراءات وأحاط بتراجم الرواة علماً عرف قدر ما سبرنا ونقحنا وصححنا، وهذا علم أهمل، وباب أُغلق، وهو السبب الأعظم في ترك كثير من القراءات، والله تعالى يحفظ ما بقي.

وإذا كان صحة السند من أركان القراءة كما تقدم تعين أن يُعرف حال رجال القراءات كما يعرف أحوال رجال الحديث».

قال الحافظ ابن الجوزي:

«واعلم أن في ذكر السير والتواريخ فوائد كثيرة، منها أنه إذا ذكرت سيرة حازم ووصفت عاقبة حاله، أفادت حسن التدبير، واستعمال الحزم، أو إن ذكرت سيرة مفرط

﴿ ١٨ ﴾

ووصفت عاقبته، أفادت الخوف من التفريط، فيتأدب المتسلط، ويعتبر المتذكر، ويتضمن ذلك شحذ صوارم المعقول، ويكون روضة للممتزه في المنقول».

وقال ياقوت الحموي في مطلع كتابه «إرشاد الأريب إلى معرفة الأديب»:

«فهذه أخبار قوم عنهم أخذ القرآن المجيد، والحديث المفيد، وبصناعتهم تُنال الإمارة، وببضاعتهم يستقيم أمر السلطان والوزارة، ويعلمهم يتم الإسلام، وباستنباطهم يُعرف الحلال من الحرام».

إن الناظر بعين البصيرة في كتب تراجم قراء مصر عبر التاريخ يجد أنها لم تحظ بتلك العناية اللائقة بها، وعلى الرغم من أن الكثير ترجم لشخصيات مصرية، فإن معظم التراجم كانت للأدباء والشعراء، وبعضها كانت لقراء المحافل والإذاعة، وذلك مع قلة الكتابات عن قراء مصر عبر التاريخ، فكان من الأولى الاهتمام بتراجم هؤلاء الأعلام، وذلك لأن قراء القرآن كان لهم المكانة العالية في كل زمان ومكان؛ لذا كانت الحاجة إلى عمل كتاب يجمع تراجم قراء مصر عبر التاريخ من الأمور الملحة، نظراً لأهميتها الكبيرة.

وقد وفقني الله عزَّ وجلَّ لعمل هذا الكتاب الذي جمعت فيه تراجم قراء مصر عبر التاريخ إلى وقتنا هذا، وبيّنت فيه أسانيد القراء قدر المستطاع، كما بيّنت فيه جميع الأسماء التي وردت بالكنى في كتاب غاية النهاية للإمام ابن الجزري، وذلك بعد تنقيح هذه التراجم من الأخطاء، وإعادة صياغتها، وزيادة مصادر كثيرة لها، وقد ترجمت لأكثر من تسعين قارئ في هذا الكتاب، وذلك في الفترة التي كانت ما قبل وفاة الإمام محمد ابن الجزري لم يتعرض لها ابن الجزري في كتابه الممتع غاية النهاية.

وقد سميت هذا الكتاب: «فريدة الدهر في طبقات قراء مصر».

وأتقدم بالشكر الجزيل: لفضيلة الشيخ أ.د/ أحمد عيسى المعصراني، وفضيلة الشيخ د/ إيهاب فكري، وفضيلة الشيخ أ.د/ وليد إدريس المنيسي، وفضيلة الشيخ د/ أيمن أحمد سعيد، وفضيلة الشيخ/ حسن مصطفى الوراق، وفضيلة الشيخ د/ محمد سكر

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السكندري، وفضيلة الشيخ/ عدنان العرضي، وفضيلة الشيخ/ هشام عبد الباري، وفضيلة الشيخ د/ محمود شمس، وفضيلة الشيخ د/ خالد حسن أبو الجود، وفضيلة الشيخ/ صابر عبد الحكم، وفضيلة الشيخ/ إبراهيم عبدالقادر أمين الوزان، وفضيلة الشيخ د/ مصطفى الحلوس، وفضيلة الشيخ د/ يحيى الغوثاني، وفضيلة الشيخ/ مصطفى شعبان الوراق، وفضيلة الشيخ د/ حامد أكرم البخاري، وفضيلة الشيخ/ عبد الحميد فياض، والشيخ د. سعيد أبو حمزة الذي قام بمراجعة الكتاب وتدقيقه.

والله أسأل أن يكون هذا العمل خالصاً لوجهه الكريم ونافعاً لجميع المسلمين،
وصلّى الله وسلم وبارك على نبينا محمد وعلى آله وصحبه الطيبين الطاهرين.

Figure 3: An Arabic biographical work on Egypt's Qur'an reciters, Volume 1.

Metric	Value
Word Error Rate	34.74%
Character Error Rate	77.68%
Overall Similarity	22.65%

This is where Qwen’s OCR clearly weakens, and it weakens in more than one way.

On ordinary paragraph text, it’s readable but noticeably rougher than English. Correct passages, like الحمد لله الذي هدانا إلى الإسلام, come through exactly right. But scattered throughout, diacritics (*tashkīl*) get dropped or altered, and words get swapped for similar-looking ones: وسير read as وسير; أحوال read as أخبار. There’s also at least one case of outright invention on an otherwise ordinary line: the word قصص (“narrated”) came back as قص القرآن — the model added a word, “the Qur’an,” that isn’t in the source at all.

On two specific pages, it broke down more seriously:

- The opening *Bismillah* invocation, بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ, came back as يُسَنِّئُ اللَّهُ الْخَزْرَجَ — not a misreading of those words, but different words entirely.
- Further down the same page, a Qur’anic citation — Surah Yūsuf 12:111, لَقَدْ كَانَ فِي قَصَصِهِمْ) عِبْرَةً لِأُولِي الْأَلْبَابِ) — was replaced with different Qur’anic-sounding phrasing altogether:) قَدْ كَذَبَ الَّذِينَ فِي قُلُوبِهِمْ مَّرَضٌ لَا يُؤْمِنُونَ) , echoing the wording of other, unrelated verses about hypocrisy and disbelief. This is not a misreading of the printed characters — it’s the model substituting in different memorized religious text instead of reading what was actually on the page.
- A dense, small-font credits page came back with every word’s Arabic letters in **reversed order** (e.g. الشيخ rendered as its letters back-to-front). This was confirmed to be a consistent, reproducible failure — rerunning that exact page twice at a deterministic setting produced identical, equally reversed output both times. This isn’t the model guessing wrong; it’s consistently misreading the script’s letter order on that specific layout.

Put together, Arabic accuracy has two separate problems: ordinary transcription noise (diacritics, similar-looking words, occasional small insertions), and outright hallucination on top of it — including, in one case, substituting in real but incorrect Qur’anic text that was never on the page.

6 The Takeaway

Qwen3.5 is a genuinely strong OCR reader for English on a machine that can’t run anything bigger — close to flawless on clean modern print, and still solidly usable on older, archaic typesetting, with the caveat that it will occasionally smooth out awkward phrasing into something more modern instead of transcribing it exactly. That caveat matters: this model does not just misread text, it sometimes *rewrites* it to sound more natural, which is a fundamentally different and harder-to-catch kind of error than a simple typo.

Arabic is where that risk becomes serious. It’s not just that the error rate roughly triples — it’s that some of those errors are outright fabrications rather than misreadings, up to and including swapping in different memorized religious phrasing in place of the Qur’anic verse actually printed on the page. For a casual transcription job, Qwen3.5’s Arabic OCR is a reasonable starting point that will still need a careful human pass. For anything where getting the exact text right actually matters — religious texts chief among them — it cannot be trusted unsupervised, full stop.